

Day 3: BIOS and POST Troubleshooting Theory

Task 1 : List 5 common BIOS or POST error messages you might see when starting a PC and write what hardware or configuration issue each one usually points to.

No.	BIOS/POST Error Message	Hardware or Configuration Issue
1.	CMOS Checksum Error	Usually indicates a weak/dead CMOS battery , corrupted BIOS settings, or incorrect CMOS configuration.
2.	Keyboard Error / Keyboard Not Detected	Usually points to a disconnected, faulty, or incorrectly connected keyboard or a problem with the keyboard port.
3.	No Boot Device Found / Boot Device Not Found	Usually indicates that the HDD/SSD is not detected , the boot drive is faulty, or the BIOS boot order is incorrect.
4.	Memory Error / Memory Size Mismatch	Usually points to a loose, incompatible, or faulty RAM module , or incorrectly configured memory.
5.	CPU Fan Error	Usually indicates that the CPU fan is not detected, is disconnected, or is not spinning properly , which can cause CPU overheating.

Task 2 : Given this scenario: On boot, a PC beeps 3 times in a repeating pattern and does not display anything on the screen. Research and write what this POST beep code typically means for an AMI BIOS, and suggest two troubleshooting steps.

Scenario

A PC is switched on, but it produces **3 beeps in a repeating pattern** and there is **no display on the monitor**.

Meaning of the Beep Code

For a traditional **AMI BIOS**, **3 short beeps indicate a base memory failure**, specifically a failure during the read/write test of the first 64 KB of system memory. In practical troubleshooting, this usually points toward a **RAM-related problem**, such as a loose, faulty, incompatible, or poorly seated memory module.

Troubleshooting Steps

1. Reseat the RAM

1. Turn off the PC and disconnect the power cable.
2. Open the computer case and locate the RAM modules.
3. Remove the RAM carefully from its DIMM slot.
4. Check the module and slot for visible dust or damage.
5. Reinsert the RAM firmly until the retaining clips lock into place.
6. Start the PC again and check whether the beep pattern and no-display problem are resolved.

Reseating the memory is a standard first troubleshooting step for this type of AMI memory error.

2. Test the RAM Modules Individually

If reseating does not solve the problem, test the RAM **one module at a time**.

- Remove all RAM modules except one.
- Try booting the PC.
- If there is still no display, move the same module to another compatible DIMM slot.
- Repeat the process with the other RAM module, if more than one is installed.

This can help determine whether a particular RAM module or DIMM slot is causing the problem. Testing with a known-good compatible memory module is another useful confirmation step.

Conclusion

The repeating **3-beep pattern on a traditional AMI BIOS generally indicates a memory/RAM failure**. The first troubleshooting actions should be to **reseat the RAM and then test the modules individually**. If the problem continues even with known-good RAM, the motherboard's memory circuitry or another related component may require further diagnosis.

Task 3 : Access the BIOS setup utility on any available PC or virtual machine, take a screenshot of the main BIOS screen, and identify the motherboard manufacturer and BIOS version.

Objective

To access the BIOS/UEFI Setup Utility and identify the motherboard manufacturer and installed BIOS version.

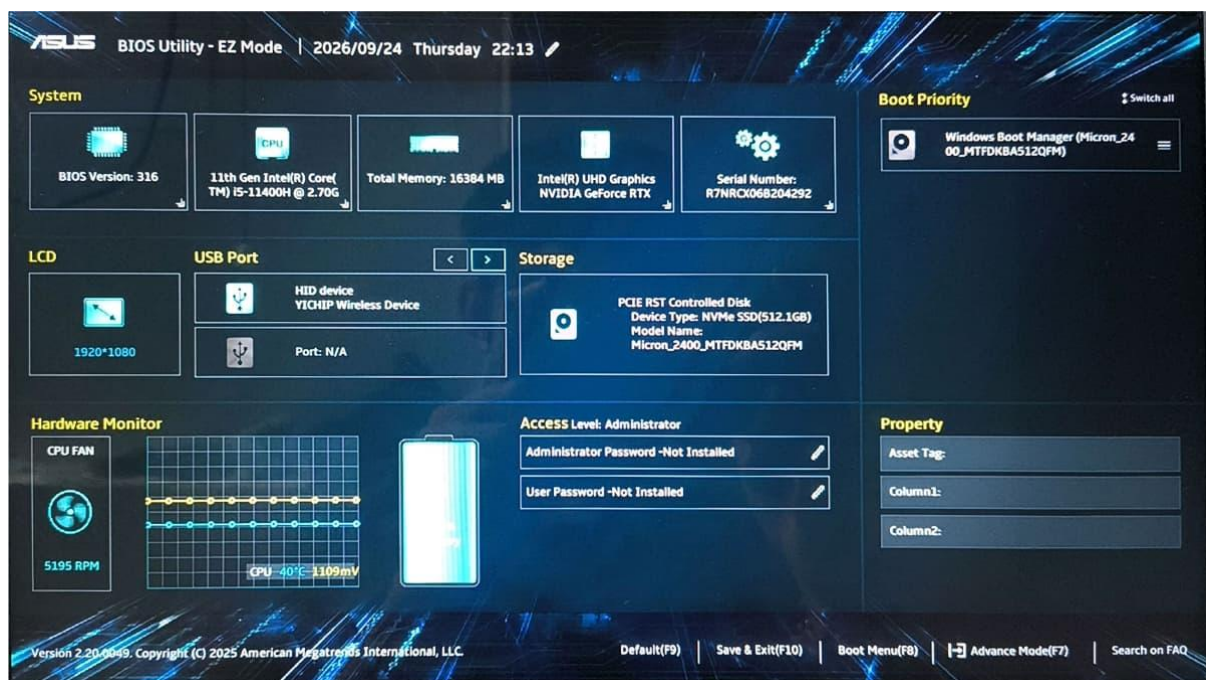
Procedure

The laptop was restarted and the BIOS/UEFI Setup Utility was opened during system startup. The main BIOS information screen was checked to identify the manufacturer and BIOS version.

BIOS Information

Information	Details
Motherboard Manufacturer	ASUS
BIOS Version	316
BIOS Date	Not displayed on the main BIOS screen
BIOS/UEFI Mode	UEFI – ASUS BIOS Utility EZ Mode

Screenshot



Task 4 : You are given a screenshot of a POST error message that says 'CMOS checksum error – Defaults loaded.' Explain what this means and describe how you would resolve this issue in a real system.

Hint: Think about the CMOS battery and BIOS settings reset.

Error Message

“CMOS checksum error – Defaults loaded.”

Meaning

This error means that the BIOS has detected that the **stored CMOS/BIOS settings are incorrect, corrupted, or have been reset**. The CMOS stores important BIOS configuration information such as the system date and time, boot settings, and hardware configuration.

A common cause is a **weak or dead CMOS battery**, usually a CR2032 battery on desktop motherboards. When the battery becomes weak, the BIOS may lose its stored settings when the computer is powered off.

How to Resolve the Issue

1. **Enter the BIOS/UEFI Setup** when the PC starts, usually by pressing **Delete, F2, or the appropriate BIOS key**.
2. Check whether the **date and time** have been reset to an incorrect value.
3. Select **Load Optimized Defaults/Setup Defaults** if required.
4. Save the BIOS settings and restart the computer.
5. If the error appears again after restarting or after the PC has been powered off, **replace the CMOS battery** with a compatible CR2032 battery.
6. After replacing the battery, enter BIOS again and set the **correct date, time, boot order, and any required custom BIOS settings**.
7. Save the settings and restart the system to confirm that the error no longer appears.

Conclusion

A **CMOS checksum error** generally indicates that the BIOS configuration stored in CMOS is invalid or has been lost. Loading the default BIOS settings can temporarily resolve it, while **replacing a weak CMOS battery** is a common permanent solution when the error repeatedly returns.